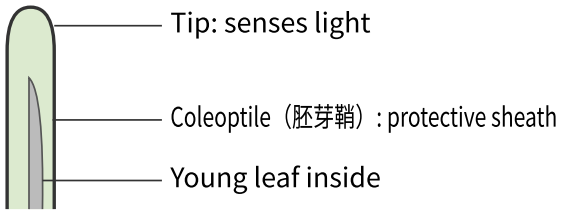
Name:_____ Class:_____ Student number:_____

Light sensing and growth in plants

Read A, then complete B. Read C only after finishing B. You do not need to operate a virtual lab.

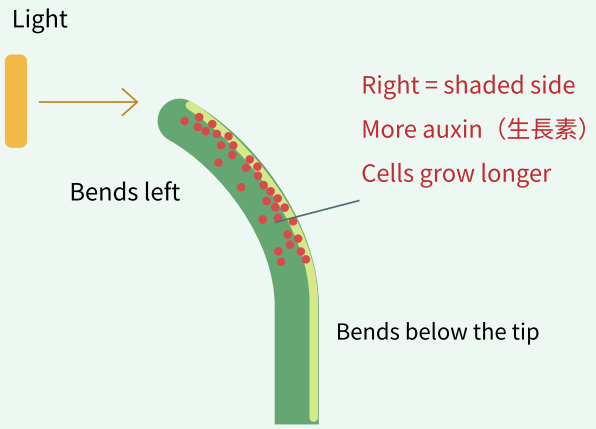


An oat coleoptile（胚芽鞘） is a sheath that surrounds and protects the young leaf. This activity examines elongation（延長） and bending of the sheath.

Learn the core concepts first

1. Directional plant growth in response to light direction is called **"phototropism（向光性）"**; growth towards light is called "positive phototropism（向光性）".
2. In this oat coleoptile（胚芽鞘） model, **Tip** helps **sense light from one side**.
3. Other research helps us understand the substance called "substance X" in this inquiry as **Auxin（生長素）**. Auxin（生長素） is a hormone that affects plant growth; the tip can produce a growth-promoting effect that **passes downwards**.
4. With light from one side, **more auxin（生長素） is found on the shaded side, so cells on that side elongate（延長） more, making the coleoptile（胚芽鞘） bend towards light**.

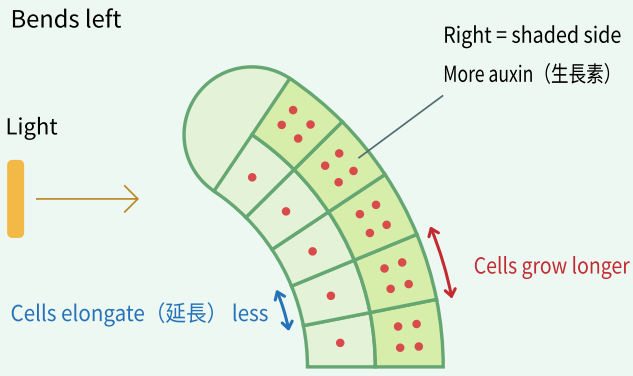
1 • Light from the left



Tip senses light → growth signal passes down

- Red dots = auxin（生長素）

2 • Light from the left: enlarged cells



More elongation（延長） on the shaded side → bends left

Worked example: a reasoned prediction

Given prediction: ‘The coleoptile（胚芽鞘） may still bend towards light when light is blocked below the tip.’ This is reasonable: the intact tip can sense light from one side and a growth-promoting effect can pass down. The lower region need not receive light directly to elongate（延長） unequally. Compare A and C to test it.

Example: identify variables

A–C changes only the shading treatment and records bending. Keep light direction and intensity, species, initial size and condition, time, temperature and water the same. The independent variable is deliberately changed; the dependent variable is the response observed or measured; controlled variables are kept the same.

Tip removal and agar（瓊脂）: apply the concepts to new comparisons

The tip senses light and can produce an effect that promotes elongation（延長） below it. A prediction that tip removal cannot affect elongation（延長） overlooks this role; check the A/D data. Cutting also creates a wound and removes tissue, so this comparison alone cannot prove a unique mechanism.

Blank agar（瓊脂） has not touched the tip; treated agar（瓊脂） has touched the tip. While agar（瓊脂） is in contact with a coleoptile（胚芽鞘） tip, a growth-promoting substance can diffuse from the tip into the agar（瓊脂）. Later, when this agar（瓊脂） is placed on a coleoptile（胚芽鞘） with its tip removed, the growth-promoting substance can diffuse out of the agar（瓊脂） and into the coleoptile（胚芽鞘）. Comparisons of treatment and placement occur in darkness; bending cannot be attributed to light acting during incubation.



Method example only: 20°, right

Worked example: angle and direction are separate

Use the vertical line as 0°. Read how far the tip direction differs from vertical. This example is 20° to the right. The same tilt to the left is still 20°, but its direction is left.

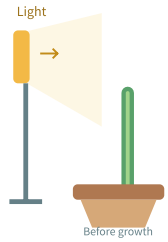
The solid pointer follows the tip direction; the dashed line is the vertical reference. This diagram teaches reading, and is not experimental data for E–H.

The 24-hour period and angles are teaching-model settings, not real constants. Elongation（延長） and bending do not directly measure auxin（生長素） concentration or identify a chemical substance. Auxin（生長素） identity and its distribution under light are understood together with other research.

1. Complete the treatment labels

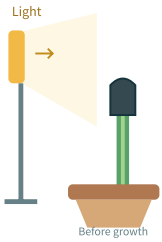
Choose labels: uncovered, tip shaded, shaded below the tip. All tips are intact and the light is on the left.

A



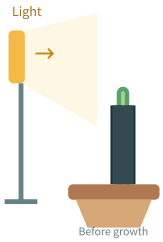
Treatment label:

B



Treatment label:

C



Treatment label:

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

2. Variables and assumptions in the main comparison

Independent variable(the factor changed on purpose)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Dependent variable(the result measured or observed)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Controlled variables(the factors kept the same)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

03 • Experiment assumptions

What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

- ☐ The coleoptiles（胚芽鞘） start in similar growth conditions.
- ☐ The light-blocking cap and sleeve do not limit elongation（延長） or bending.
- ☐ The treatments do not cause clear temperature differences between setups.
- ☐ If the plants are the same type, their starting size does not need to be controlled.

3. What baseline does uncovered A provide? What alone changes in A–B and A–C? Give at least two conditions that must be the same.

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

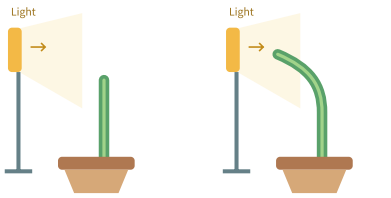
4. Observe and analyse shading comparisons

Light from the same side, three light-blocking treatments

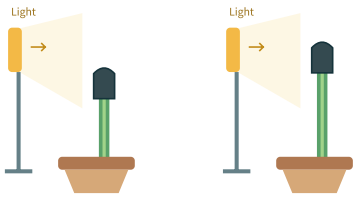
The light is on the left for every setup. Before treatment, the coleoptiles（胚芽鞘） are similar in type, starting height and growth stage. Light intensity, exposure time, temperature and water supply are the same. The opaque（不透光） sleeve covers the whole part below the tip, leaving the tip exposed. The sleeve can bend as the plant grows and does not limit elongation（延長）.

The animation speeds up growth and shows a typical response. The time and data are teaching simulations, not real measurements. The model does not show differences between individual plants.

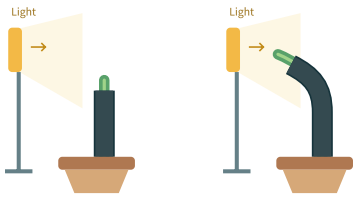
A • No light blocked



B • Light blocked at the tip



C • Light blocked below the tip



Setup	Treatment	Bending direction
A	No light blocked	（Bends left ／ Bends right ／ No clear bending）
B	Light blocked at the tip	（Bends left ／ Bends right ／ No clear bending）
C	Light blocked below the tip	（Bends left ／ Bends right ／ No clear bending）

Hint: compare treatments and before/after shapes, using A as the baseline.

1. Compare A and B: after **blocking light at the tip**, what changes?（No clear bending towards the light. ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend?（It can still bend towards light. ／ No clear bending towards the light.）

3. Which part may **sense light from one side**?（Tip ／ Only the lower part）

5. Compare elongation（延長） of intact A and cut-tip D

Use the same left-side light and incubation conditions; compare whether the tip is present. Use the before/after images to circle yes/no elongation（延長）, then complete the conclusion. Cutting also creates a wound; do not infer a unique mechanism.

A • No light blocked



D • Tip removed



Setup	Treatment	Elongation（延長）
A	No light blocked	（Yes ／ No）
D	Tip removed	（Yes ／ No）

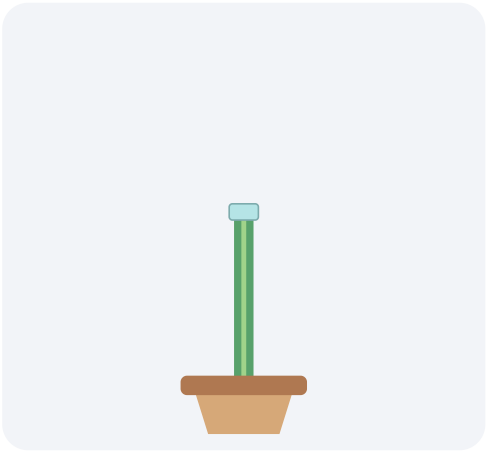
1. What does comparing elongation（延長） in A and D support?（The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）. ／ The presence of the tip does not affect the coleoptile（胚芽鞘）’s growth / elongation（延長）.）

6. Record E/F angles and directions in darkness

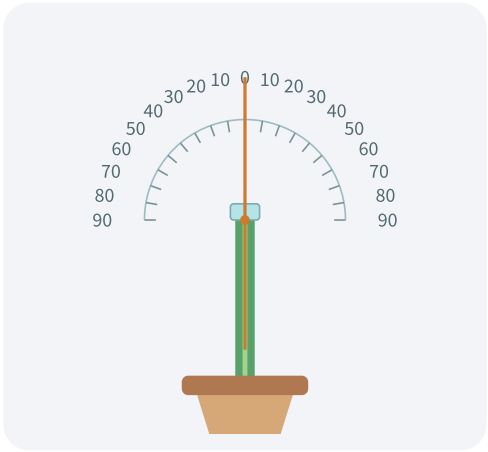
All E–H are incubated in darkness for 24 hours (a teaching model), with the same species, initial size, time, temperature and water. Use the A2 method to read final protractors: angle is the magnitude from vertical; record direction separately. Agar（瓊脂） type and position remain fixed during incubation.

Extension 2 materials: oat coleoptiles（胚芽鞘） with tips removed × 4, growing containers × 4, blank agar（瓊脂） × 1, treated agar（瓊脂） × 3, timer × 1.

E • Blank agar（瓊脂） • Centre

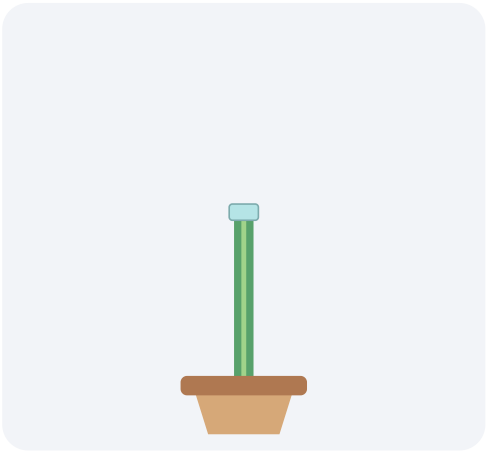


Darkness • Before growth

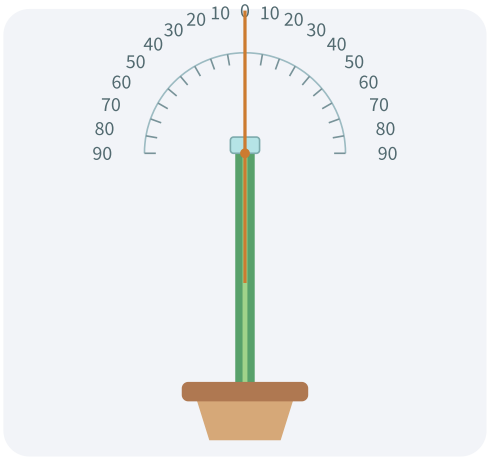


Darkness • After growth

F • Treated agar（瓊脂） • Centre



Darkness • Before growth



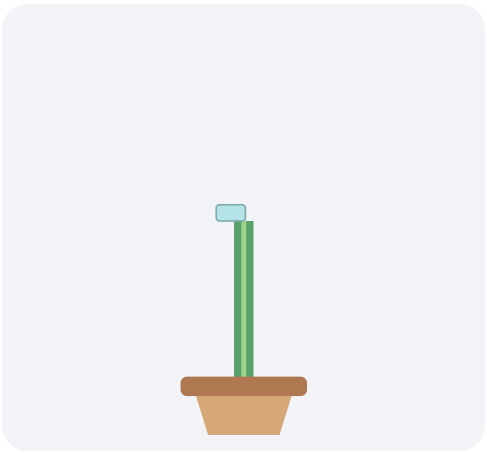
Darkness • After growth

Setup	Treatment	Angle (°)	Bending direction
E	Blank agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）
F	Treated agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）

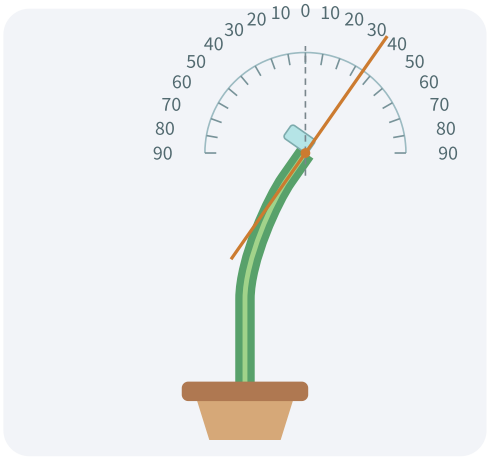
6. (continued) Record G/H and analyse E–H

G/H are also incubated in darkness. They use the same treated agar（瓊脂） at different positions. Read the angle and record direction, then answer the three questions below.

G • Treated agar（瓊脂） • Left

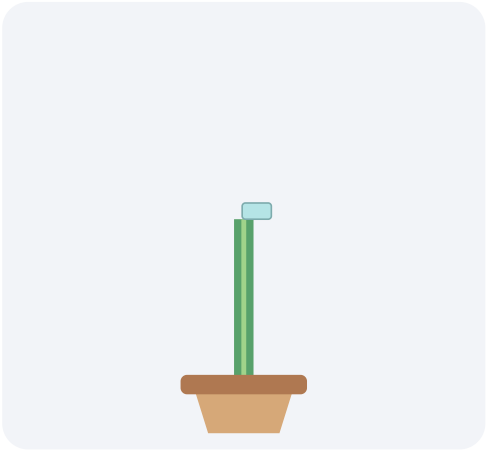


Darkness • Before growth

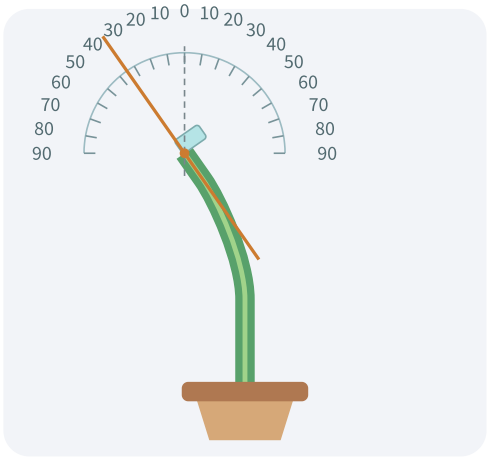


Darkness • After growth

H • Treated agar（瓊脂） • Right



Darkness • Before growth



Darkness • After growth

Setup	Treatment	Angle (°)	Bending direction
G	Treated agar（瓊脂） • Left	_____ °	（Bends left ／ Bends right ／ No clear bending）
H	Treated agar（瓊脂） • Right	_____ °	（Bends left ／ Bends right ／ No clear bending）

1. What does the comparison of E and F support?（The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support?

◯ The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

◯ The position of treated agar（瓊脂） **does not** affect the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

7. Identify independent variables in different comparisons

Circle one option in each row; distinguish variables across the investigations.

A–D:

（whether the tip is present ／ light direction）

E–F:

（whether agar（瓊脂） previously touched the tip ／ agar（瓊脂） position）

F–G–H:

（whether agar（瓊脂） previously touched the tip ／ treated agar（瓊脂） position）

8. Observe the window-side context, then apply the concepts

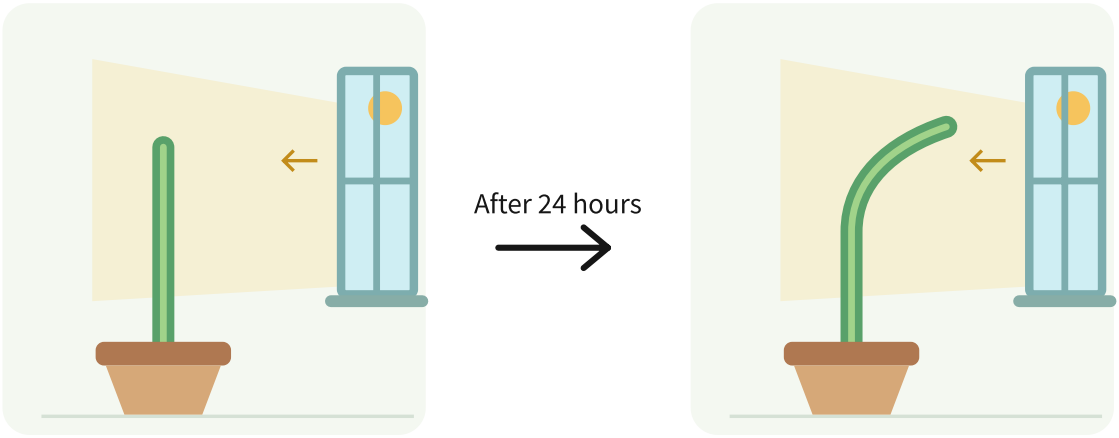


Illustration of growth changes; 24 hours is the time chosen for this situation.

Describe the change in coleoptile（胚芽鞘） shape and growth direction in the before/after images.

Use the learning points to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.

Connect light direction, the shaded side, auxin（生長素） and differential elongation（延長） of cells on the two sides.

Keep these first answers. Read C only after completing B; write corrections separately without erasing the originals.

Name:_____ Class:_____ Student number:_____

Read only after completing B. Keep A available; check these answers without overwriting your first responses.

B1: labels, variables and fair comparisons

Labels: A uncovered; B tip shaded; C shaded below the tip.

Independent variable	Light-blocking treatment
Dependent variable	The coleoptile（胚芽鞘）’s bending response
Controlled variables	Light direction and intensity; Coleoptile（胚芽鞘） type, starting height and growth stage; Time exposed to light; Temperature and water supply

Valid assumptions: similar initial growth condition; shading does not restrict elongation（延長） or bending; shading creates no clear temperature difference. The same species does not remove the need to control initial size.

A is uncovered and provides a baseline without shading in the same environment. A–B changes tip shading; A–C changes shading below the tip. Keep conditions such as initial size and temperature the same to avoid other influences.

B2 / B3 / B4: check observations

Setup	Treatment	Direction
A	No light blocked	Bends left
B	Light blocked at the tip	No clear bending
C	Light blocked below the tip	Bends left

Setup	Treatment	Elongation（延長）
A	No light blocked	Yes
D	Tip removed	No

Setup	Treatment	Angle	Direction
E	Blank agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
F	Treated agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
G	Treated agar（瓊脂） • Left	35°（Teaching model; ±2°）	Bends right
H	Treated agar（瓊脂） • Right	35°（Teaching model; ±2°）	Bends left

Method check: E/F are 0°. G/H accept 33–37°, with equal angle magnitudes but opposite directions. Observations and readings are model data, not auxin（生長素） concentrations.

B2 / B3 / B4: check analysis

3. Which part may sense light from one side? Tip

1. Compare A and B: What changes when light is blocked at the tip? No clear bending towards the light.

2. Compare A and C: Does it still bend when the lower part receives no direct light? It can still bend towards light.

1. What does comparing elongation（延長） in A and D support? The presence of the tip also affects the coleoptile（胚芽鞘）' s growth / elongation（延長）.

1. What does the comparison of E and F support? The tip may produce a growth-promoting effect that can be transferred.

2. What does the comparison of F, G and H support? The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

B5: independent application

A–D: whether the tip is present. E–F: whether agar（瓊脂） previously touched the tip, with both in the centre. F–G–H: treated agar（瓊脂） position, with the same agar（瓊脂） type. The main comparison changes shading treatment.

Window-side context: initially straighter, then bent towards the light on the right.

The light is on the right, so the left is the shaded side. The tip senses light from one side; auxin（生長素） is more abundant on the shaded left and its growth-promoting effect passes down. Cells on the left elongate（延長） more than those on the right, making the coleoptile（胚芽鞘） bend towards the right-side light.

Cutting creates a wound and removes tissue, so it cannot establish a unique mechanism alone. Dark agar（瓊脂） results support a transferable growth-promoting effect and unequal elongation（延長）. Bending is not caused by light during incubation; the results neither measure auxin（生長素） concentration directly nor identify the substance.

Check and correct

Identify one answer you corrected and the image or comparison supporting the correction. If correct initially, give one supporting piece of evidence.

In one sentence, explain how auxin（生長素）, differential elongation（延長） and bending are related.

Research conditions and distribution

Experimental group: IBL + VL. Comparison group: no prelesson lecture and no VL operation; students self-study using A/B worksheets. Paper follows explanation → worked examples → guided practice → independent application → checking/correction, with shared content and reasonable science-process practice rather than the VL's delayed concept reveal.

Stage	15 min	20 min	Distribution and support
A: concepts and images	3	4	Give A at the start; no prelesson lecture.
A: worked examples	2	2	Students read independently.
B1–B4: guided practice	5	7	Give B; use the shared printed hints.
B5: independent application	2	3	A remains available; no individual answers.
C: check and correct	3	4	Give C only after B is finished and first answers retained.

Choose one common 15–20-minute total for both groups, including reading, answering and feedback. These are two planning schedules, not measured durations. Record actual time and deviations; paper actions or completion are not VL abilities.

Teacher clarification and feedback

During A/B, clarify only procedures, vocabulary and diagram-reading conventions using the same planned hints. Do not add a lecture or give individual answers or growth directions. Record extra conceptual teaching as a protocol deviation. Retain completed B, then give C and allow content feedback based on C; record corrections separately. D and the full pack are for teachers only.

Shared goals, separate scoring and outcomes

Share the context, three investigations, model images, core concepts and goals in observing, variables, fair comparisons, measurement, inference and communication; teaching sequence differs. Paper does not score advance guesses, clicks, demonstrations, animation completion or virtual dragging. D's 14 points are formative and must not be compared directly with VL's 32. Keep common SPSAI/CKT pre/post-tests separate and unchanged; do not add experimental-only opinion or interview items.

VL1 informs PDF interface and layout only. Use displayed C for below-tip shading and D for cutting; keep legacy C/D records and compatibility unchanged. Generation, preview and download do not create learner records or write answers, cloud data, events or teacher Excel.

Student checking answers

C1/C2 provide all closed-question answers and key points for open responses in B; the complete pack includes both C pages. For teacher judgement, accept scientifically equivalent wording rather than exact copying.

Observation, labels, variables and controls

P1: straighter initially, then bends towards the light on the right. P2: A uncovered, B tip shaded, C shaded below the tip. P3: shading is IV; bending is DV; light direction/intensity, species/initial size/condition, time, temperature/water are CVs. The first three assumptions apply; same species does not remove size control.

A is uncovered and provides a baseline without shading in the same environment. A–B changes tip shading; A–C changes shading below the tip. Keep conditions such as initial size and temperature the same to avoid other influences.

Data and conclusions from all three investigations

Main comparison: bending direction

A	No light blocked	Bends left
B	Light blocked at the tip	No clear bending
C	Light blocked below the tip	Bends left

Extension one: presence or absence of elongation（延長）

A	No light blocked	Yes
D	Tip removed	No

Extension two: angle and direction

E	Blank agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
F	Treated agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
G	Treated agar（瓊脂） • Left	35°（Teaching model; ±2°）	Bends right
H	Treated agar（瓊脂） • Right	35°（Teaching model; ±2°）	Bends left

3. Which part may sense light from one side? Tip

1. Compare A and B: What changes when light is blocked at the tip? No clear bending towards the light.

2. Compare A and C: Does it still bend when the lower part receives no direct light? It can still bend towards light.

1. What does comparing elongation（延長） in A and D support? The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）.

1. What does the comparison of E and F support? The tip may produce a growth-promoting effect that can be transferred.

2. What does the comparison of F, G and H support? The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

A–D: whether the tip is present. E–F: whether agar（瓊脂） previously touched the tip, with both in the centre. F–G–H: treated agar（瓊脂） position, with the same agar（瓊脂） type. The main comparison changes shading treatment.

Reflection and correction

Right-side light makes the left the shaded side. The tip senses light; more auxin（生長素） on the left promotes greater left-cell elongation（延長）, so the coleoptile（胚芽鞘） bends towards the light on the right.

C2 correction: retain first answers, identify a correction and one data-based reason. If initially correct, provide supporting evidence. The causal chain must show greater elongation（延長） on the shaded side, not the lit side.

Cutting creates a wound and removes tissue, so it cannot establish a unique mechanism alone. Dark agar（瓊脂） results support a transferable growth-promoting effect and unequal elongation（延長）. Bending is not caused by light during incubation; the results neither measure auxin（生長素） concentration directly nor identify the substance.

VL-to-paper question mapping

Original / ID	Treatment	New question	Shared goal	Reason
P1 • Initial shape and direction observation	Retain / static	Describe changes in shape and direction in the window-side images.	Observation and recording	Same context; images replace animation.
E1 • Original main-inquiry hypothesis and reason	Worked example	Evaluate a given prediction: blocking light below the tip may still allow bending towards light.	Testable prediction and reasoning	Teach the mechanism first; model an evidence-based prediction without scoring an original hypothesis.
P2 • Autonomous setup drawing / written design	Rewrite / static	Label the shading treatments in three provided diagrams.	Identify apparatus and treatments	Provide the same setups for scientific labelling; do not equate this with virtual operation.
P3 • Main-inquiry variables and assumptions	Retain	Select main-inquiry variables and all applicable assumptions.	Classify and control variables	Keep variable choices and assumptions about initial size, temperature and unrestricted growth.
P4 • Autonomous control design	Equivalent rewrite	Explain the role of uncovered A and fair comparisons A–B / A–C.	Fair comparison and controls	Supply the control while retaining reasoning about baseline and controlled conditions.
P5 • Main A / B / C direction records	Static	Circle the bending direction after incubation for each setup.	Observation and recording	Shared before/after models and original direction options.
P6 • Compare A and B: what changes when the tip is shaded?	Retain	Compare A and B: what changes when the tip is shaded?	Comparison and inference	Keep the question and options; apply taught concepts to data.
P7 • Compare A and C: does it bend without direct light below the tip?	Retain	Compare A and C: does it bend without direct light below the tip?	Comparison and inference	Use displayed C without renumbering legacy D records.
P8 • Which part may sense light from one side?	Retain	Which part may sense light from one side?	Infer light sensing from evidence	Keep the A/B and A/C evidence chain.
E2 • Original extension-one prediction and reason	Application example	Use the tip's growth-promoting role to evaluate a cut-tip prediction and its limits.	Reasoned prediction	Teach the role and wound caveat rather than scoring an advance guess.

VL-to-paper question mapping

Original / ID	Treatment	New question	Shared goal	Reason
P9 • A/D elongation（延長） records and conclusion	Retain / static	Record whether A and D elongate（延長）, then conclude about the tip's role.	Qualitative observation and inference	Shared intact A / cut-tip D images and yes/no options.
E3 • Original extension-two prediction and reason	Explanation	Explain how agar（瓊脂） carries an effect and how placement relates to differential elongation（延長）.	Model understanding and prediction	Teach the method before independent interpretation of dark E–H.
P10 • E–H adjustable-pointer measurement and direction	Static	Read static protractors and record direction separately.	Measurement and recording	Retain the 0/0/35/35 model; image reading is not virtual-operation scoring.
P11 • What does comparing E and F support?	Retain	What does comparing E and F support?	Evidence for a transferable effect	Both are central; only prior tip contact differs.
P12 • What does comparing F, G and H support?	Retain	What does comparing F, G and H support?	Placement and bending	Same treated agar（瓊脂）, different positions; no light during dark incubation.
P13 • Extension-two cloze on differential elongation（延長）	Retain	Select more/less X on the left, more/less elongation（延長）, and left/right bending.	Reason about differential elongation（延長）	Keep options as circling choices rather than dropdowns.
P14 • Fair-comparison aims across three investigations	Equivalent application	Identify the independent variable separately for A–D, E–F and F–G–H.	Transfer variable classification	Distinguish shading, tip removal, agar（瓊脂） conditioning and placement.
P15 • Use the learning points to explain the right-bending coleoptile（胚芽鞘） in the diagram.	Retain / independent application	Use the learning points to explain the right-bending coleoptile（胚芽鞘） in the diagram.	Integrate concepts in an explanation	Link right-side light, shaded left, auxin（生長素） and differential elongation（延長）, not hypothesis revision.
C1 • Learning points revealed at the end	Move to A; C checks/corrects	Read concepts from the start; after B, check answers and record corrections separately.	Conceptual understanding and correction	Fits the self-study condition without a prelesson lecture.

Paper formative assessment (separate from VL)	Points	Paper evidence and criteria
Observation and recording	0–2	B2/B3/B4/B5: direction, presence/absence of A/D elongation（延長） and window-side shape. 2: accurate and complete; 1: partly accurate; 0: no valid record.
Variable classification	0–2	B1/B5: main shading IV/DV/CVs and independent variables for cutting, agar（瓊脂） treatment and placement. 2: correct across comparisons; 1: partly correct; 0: not identified.
Fair comparison and assumptions	0–2	B1: labels, A as baseline, initial size and temperature controls, unrestricted growth under shading. 2: sound and complete; 1: partial; 0: no fair-comparison understanding.
Evidence-based inference	0–2	B2/B3: conclusions from A/B, A/C, A/D, E/F and F/G/H. 2: distinguishes the effects each comparison supports; 1: partial; 0: inconsistent with data.
Static protractor reading	0–2	B3/B4: E/F require 0°; G/H accept 33–37°, 0.5 per setup. Assess image reading only, not dragging or virtual-operation skill.
Scientific communication and correction	0–2	B1/C2: fair-comparison reasoning and evidence for separately recorded corrections. 2: clear reason–evidence link; 1: partial; 0: no valid link. Keep B first answers; assess C corrections separately.
Concept application	0–2	B4/B5: left agar（瓊脂） → more X → greater left elongation（延長） → right bend; right light → shaded left → more auxin（生長素） → greater left elongation（延長） → right bend. 2: both causal chains correct; 1: partial; 0: reversed or unexplained.
Total and use	14	For paper formative feedback only. Not the VL 32-point score, original-prediction or completion score, and not an outcome test. Evaluate research outcomes with the approved common SPSAI/CKT assessments.